

Herald



Herald — ATMOSphere Workable-Items Integration (Operator Guide)

Field	Value
Revision	1
Created	2026-05-29
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Status	active
Status summary	Operator-facing reference for the ATMOSphere ^[?] Herald workable-items integration (Phase 2 build). Documents the OUTBOUND flow (workable-items SQLite SSoT + Markdown trackers → commons_watch → commons_workable.Diff → pherald/internal/workflow bridge → existing pherald/internal/runner fan-out → channels) and the INBOUND flow (operator message → Claude Code dispatch → <<<HERALD-REPLY>>> action router → ItemMutator / investigation-with-confirmation). Grounded in the master plan ~/Documents/ATMOSphere_Herald_Integration_Plan.md. ANTI-BLUFF: built-and-tested pieces are marked LIVE; not-yet-built or externally-gated pieces are marked PLANNED / PENDING and never implied to work.
Issues	HRD-150, HRD-151 (formatter polish), HRD-154, HRD-155, HRD-156, HRD-157, HRD-158, HRD-131
Issues summary	DB materialization + MD ^[?] SQLite regenerator + preference routing + full-automation test suite + ATMOSphere registration + covenant propagation are still open — see §1.3 + §7.
Fixed	(n/a — new guide)
Continuation	bump when docs/workable_items.db is materialized (HRD-155/HRD-131 Phase 3), when the MD ^[?] SQLite regenerator lands (HRD-150), when preference/quiet-hours routing is enforced (HRD-154), when the live MTPROTO round-trip evidence lands (HRD-156), and when Herald is registered as a tools/herald submodule in ATMOSphere (HRD-157).

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§1. Overview

§1.1 What the integration does

The ATMOSphere^[?]Herald integration connects ATMOSphere's workable-items tracking system (the project's issue/ticket lifecycle) to its operators via Herald, in two directions:

- **Outbound (watch → notify).** Herald watches the workable-items SQLite single-source-of-truth (SSoT) plus its `Issues.md` / `Fixed.md` Markdown trackers. On every change it computes a per-property diff and fans a notification out to Subscribers over channels (Telegram primary) — Jira/ClickUp-style: item creation, each property change (with the exact old→new value), status transitions, content updates, and deletions.
- **Inbound (message → act).** Operators message the channel and Herald routes the message through a Claude Code dispatch into a structured `<<<HERALD-REPLY>>>` action. Workable-item CRUD (`item.update` / `item.delete`) and investigations (`investigation.start`) are supported. Mutating actions proposed by an investigation run only after an explicit `CONFIRM <token>` reply (act-with-confirmation).

Because both directions converge on the same `commons_workable` store, an operator-driven update produces the same per-property diff notification a file edit would — Subscribers always see "what changed exactly" regardless of who changed it.

§1.2 The requirements (R1–R7) and their honest status

The master plan (`~/Documents/ATMOSphere_Herald_Integration_Plan.md` §2.3) defines seven requirements. Current Herald-side status:

Req	Capability	Status	Where
R1	Watch the SQLite SSoT (create / update / delete)	BUILT (LIVE)	<code>commons_workable</code> (store + diff change-feed) + <code>commons_watch</code> (fsnotify + WAL-poll). Tested green.
R2	Watch MD trackers + keep in sync with the SSoT	PARTIAL	The watcher watches the MD trackers (<code>commons_watch</code>), and the parser reads them (<code>commons_workable/parser.go</code>). The bidirectional MD ^[?] SQLite regenerator + drift resolution is PLANNED (HRD-150).
R3	Emit a notification per event with the exact per-property diff	BUILT (LIVE)	<code>commons_workable.Diff</code> (per-property change-feed) + <code>pherald/internal/workflow</code> (CloudEvent mapper + diff renderer). Tested green.

Req	Capability	Status	Where
R4	Subscribers receive over channels (Telegram primary)	BUILT (transport reused)	pherald/internal/runner fan-out, driven by workflow.Notifier. NOTE: full PG-backed subscriber resolution is PENDING (HRD-156); pherald watch currently fans out to the configured channel targets directly (see §4.4).
R5	Inbound → process → act (update / extend / create / investigate / return)	PARTIAL (LIVE for the implemented actions)	item.update / item.delete / investigation.start are wired in pherald/internal/inbound with act-with-confirmation. Investigation autonomy scope is an open operator decision (plan §8.6).
R6	Coherent full CRUD driven by BOTH SSoT-change (notify) and inbound (mutate)	PARTIAL	The inbound write path (ItemMutator → commons_workable.Repo) and the outbound watch path both use commons_workable. The closing seam — having an inbound mutation also regenerate the MD trackers and re-emit through the watcher in one process — depends on the regenerator (HRD-150) and the daemon co-residency wiring.
R7	Full-automation anti-bluff tests with physical evidence	PARTIAL	Hermetic unit + real-SQLite + real-fsnotify + real-dispatcher tests are green (§7). The live MTPROTO Telegram round-trip + stress/chaos + paired §1.1 mutation gate suite is PENDING (HRD-156).

§1.3 What is NOT yet done (do not assume it works)

PLANNED / PENDING, explicitly:

- **The SQLite SSoT file does not exist yet.** No docs/workable_items.db is materialized in this repo. Materialization is **PENDING** (HRD-155 + HRD-131 Phase 3). See §3.5.
 - **The MD[?]SQLite bidirectional regenerator (R2 / HRD-150) is PLANNED.** Today the parser reads the trackers and the watcher watches them, but Herald does not yet write the trackers back from the DB or resolve drift between them.
 - **PG-backed subscriber resolution** for the watch path is **PENDING** (HRD-156). pherald watch fans out to the configured channel targets, not yet to a PG Subscriber set.
 - **Preference / quiet-hours routing** (HRD-154) is **PLANNED** — the PreferenceSet / QuietHours types exist but the resolver does not yet honour them.
 - **The full-automation anti-bluff test suite** with live Telegram evidence via MTPROTO (HRD-156) is **PENDING** — it requires the operator credential bootstrap. Until then the live tests honestly SKIP-with-reason (§11.4.3), they do not PASS.
 - **ATMOSphere-side registration** (HRD-157): Herald is not yet a proper submodule of ATMOSphere (the existing ./herald gitlink is a broken orphan stub). Phase 3 replaces it with a tools/herald submodule. This is **PLANNED**.
 - **Covenant verbatim-phrase propagation** into ATMOSphere QWEN.md files (HRD-158) is **PLANNED**.
-

§2. Architecture

§2.1 Outbound — SSoT change → Subscriber notification (LIVE)

ATMOSphere workable-items

```
docs/Issues.md  
docs/Fixed.md    } edits → workable_  
                  items.db | (+ sync)  
                  (SQLite SSoT)
```

Herald (commons_workable + common

```
commons_watch.Watcher  
| fsnotify on .db + .db-wal + .  
| PLUS a WAL-poll fallback (mt  
| the -wal sidecar so the main
```

```
▼  
pherald watch — runWatch loop  
  snapshot prev := Repo.List(Issues) + Repo.  
  on every watch Event (or safety-net reconc  
  curr := re-list → commons_workable.Dif  
  prev := curr  
  | []workable.Change (per-prop
```

```
▼  
pherald/internal/workflow  
  ChangesToEvents → 1 CloudEvent per Change  
                  (type digital.vasic.her  
  RenderChange    → 1 Jira/ClickUp-style di  
  Notifier.Notify → feeds each rendered lin
```

```
▼  
pherald/internal/runner.ChannelDispatcher (   
  per-recipient commons.Channel.Send + del
```

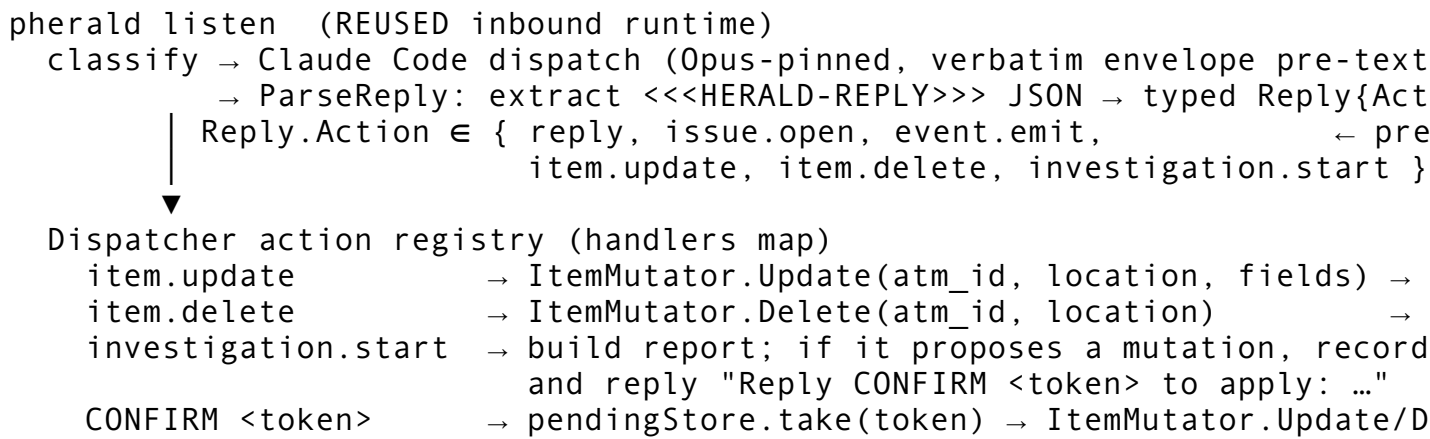
```
▼  
Telegram main group / per-subscriber channel  
"NEW ATM-238 created"  
"🔄 ATM-238 status: In progress → Ready for  
"✎ ATM-238 severity: Critical → Medium"
```

Design facts (verified in code):

- **The DB is the change-detection anchor.** Herald keeps its own prior-state snapshot keyed on the composite (`atm_id`, `current_location`) and diffs consecutive snapshots — it does not rely on the tool's `item_history` (which is not field-level and is unpopulated).
- **WAL handling is explicit.** `pherald watch` adds the `-wal` / `-shm` sidecars of every `.db` path to the watch set (`withWALSidecars`) and runs a safety-net reconcile ticker (default 1s) so a logical mutation is detected even when the main inode does not emit a timely `fsnotify` event.
- **The fan-out is reused verbatim.** `workflow.Notifier` owns no delivery logic; it drives the production `runner.ChannelDispatcher.Process` → `commons.Channel.Send` path. Only the `change→CloudEvent` producer and the diff renderer are new.

§2.2 Inbound — operator message → action → CRUD / investigation → reply (LIVE for implemented actions)

```
Operator → channel: "ATM-238 set status Ready for testing; add note: retri  
|  
▼
```



Design facts (verified in code):

- **The 3-way action switch became an extensible registry.** pherald/internal/inbound/dispatcher.go routes by Reply.Action through a handlers map; the workable actions are registered alongside the original three.
- **ItemMutator is the single inbound write surface** (item_mutator.go), parallel to the existing IssueOpener. Production binds RepoMutator over a real commons_workable.Repo; unit tests bind a recording fake. RepoMutator.Update reads the row, applies the field deltas, and writes it back through Repo.Update — a missing row surfaces as workable.ErrNotFound, an invalid status is rejected by the closed-set check, and an unknown column name is rejected loudly (no silent no-op).
- **investigation.start is act-with-confirmation** (operator decision 2026-05-29). A proposed mutation is recorded in a pendingStore under a token; the mutation runs only on a subsequent CONFIRM <token> message. The entry is consumed on lookup so a replayed CONFIRM cannot double-apply. A report-only investigation (no proposed mutation) emits no prompt and stores nothing.

§2.3 Module / layer placement

Unit	Layer	Status	Responsibility
commons_workable/	L1 foundation	BUILT	SQLite open + canonical schema, full CRUD repo, per-property diff change-feed, ATMOSphere Markdown-tracker parser
commons_watch/	L1 foundation	BUILT	fsnotify wrapper + debounce-coalesce + SQLite-WAL poll fallback
pherald/internal/workflow/	flavor-internal	BUILT	change→CloudEvent mapper, Jira/ClickUp diff renderer, Notifier over the real ChannelDispatcher
pherald/internal/inbound/ (workable extension)	flavor-internal	BUILT	action registry + item.update/ item.delete/ investigation.start + ItemMutator + pending/confirm flow

Unit	Layer	Status	Responsibility
pherald watch	flavor binary subcommand	BUILT	the watch → diff → notify daemon endpoint bidirectional sync; references
MD[?]SQLite regenerator	commons_workable + scripts	PLANNED (HRD-150)	constitution/scripts/workable-items/ per §11.4.74

§3. The SQLite single-source-of-truth

§3.1 Canonical path

The canonical DB path is `docs/workable_items.db` (no leading dot). `pherald watch` resolves it from, in order: the `--db` flag, the `HERALD_WORKABLE_DB` env var, then the `docs/workable_items.db` default.

Note: HRD-131 historically referenced `docs/.workable_items.db` (leading dot). The constitution §11.4.95 canonical path is `docs/workable_items.db` and the code uses the dot-less form; reconcile to the dot-less canonical path when the DB is materialized (HRD-131 Phase 2+).

§3.2 Schema (mirrored verbatim from ATMOSphere)

`commons_workable.Open(path)` opens (creating if absent) the DB, sets `PRAGMA journal_mode=WAL + PRAGMA foreign_keys=ON`, pins a single connection (so connection-scoped PRAGMAs hold), and applies the schema idempotently (`CREATE TABLE IF NOT EXISTS`). The driver is the pure-Go `modernc.org/sqlite` (no CGO).

Three tables:

- `items` — composite primary key (`atm_id`, `current_location`). Columns: `atm_id`, `type` (`CHECK IN ('Bug', 'Feature', 'Task')`), `status`, `severity`, `title`, `description`, `forensic_anchor`, `closure_criteria`, `composes_with` (JSON array as TEXT), `current_location` (`CHECK IN ('Issues', 'Fixed')` default 'Issues'), `body_md`, `created_at`, `last_modified`.
- `item_history` — append-only audit (`event_type IN ('Opened', 'Updated', 'Reopened', 'Fixed', 'Implemented', 'Completed', 'Observed')`, `by IN ('AI', 'User')`, `on_date`, `reason`, `evidence_path`, `created_at`). NOTE: this table is schema-defined but Herald does NOT rely on it for diffs — it is not field-level and is currently unpopulated. Herald computes diffs from its own prior-state snapshot.
- `meta` — key/value with `last_modified`.

§3.3 The 10-value status closed set

`commons_workable.StatusValues` is the canonical closed set; Create/Update reject any status outside it (no silent acceptance):

Queued
In progress
Ready for testing

In testing
Reopened
Operator-blocked
Fixed (→ Fixed.md)
Implemented (→ Fixed.md)
Completed (→ Fixed.md)
Obsolete (→ Fixed.md)

Types are the closed set Bug | Feature | Task (enforced by the schema CHECK).

§3.4 The parser and the format match

`commons_workable.ParseTracker(markdown, location)` reads ATMOSphere's real tracker format directly — it does NOT require the `## ABC-123 — title` shape the constitution tool's parser expects. It accepts H2 headings in shapes like:

```
## §GL CRITICAL — [ATM-238] Netflix login failure on D3
## SYS — [ATM-101] Disk pressure alerting
## §UX — Tidy the onboarding copy
## A. Global blockers (section header — skipped)
```

Rules: an item is an H2 block whose body contains a `**Status:**` line; a heading with no `**Status:**` is treated as a section header and skipped. The `[ATM-NNN]` bracket is the id; a bracket-less item heading gets a stable derived id (`ATM-DERIVED-<8hex>` from a sha1 of the heading). `**Status:** / **Type:** / **Severity:**` metadata lines populate those fields; the raw body block becomes `body_md`.

§3.5 Relationship to the trackers and the constitution tool — PENDING materialization

- The Markdown trackers (`docs/Issues.md / docs/Fixed.md`) are today the live source; the **DB does not exist yet** (no `docs/workable_items.db`).
- Materialization is **PENDING (HRD-155 + HRD-131 Phase 3)**: build/operate the constitution workable-items tool (`constitution/scripts/workable-items/`), supply the ATMOSphere-format parser the tool lacks, run `sync md-to-db` against the real trackers, `validate`, and commit the DB per §11.4.95 (version-controlled SSoT; only the `-wal/-shm` sidecars gitignored).
- Per §11.4.74 (catalogue-first), Herald references the constitution tool for the implemented `sync / diff / validate` rather than reimplementing them; the regenerator + drift resolution is HRD-150.

§4. Running pherald watch

Prerequisite: a materialized `docs/workable_items.db`. Until HRD-155/HRD-131 Phase 3 land, the DB does not exist, so a real run depends on that PENDING work. `pherald watch` will create an empty schema-only DB if the path is absent, but it will have no items to diff.

§4.1 Command

`pherald watch [flags]`

Long-running. It (1) opens the SSoT and snapshots every item at the watched locations (`Issues` + `Fixed`), (2) starts a `commons_watch.Watcher` over the DB file (+ WAL sidecars) and the trackers, (3) on every change re-lists, diffs against the prior snapshot, renders each per-property delta, and fans it out through the production `ChannelDispatcher`. `SIGINT/SIGTERM` cancels the loop cleanly.

§4.2 Flags

Flag	Default	Meaning
<code>--db <path></code>	<code>\$HERALD_WORKABLE_DB</code> , else <code>docs/workable_items.db</code>	Workable-items SQLite DB path
<code>--issues <path></code>	<code>docs/Issues.md</code>	Issues.md tracker path (watched)
<code>--fixed <path></code>	<code>docs/Fixed.md</code>	Fixed.md tracker path (watched)
<code>--poll <duration></code>	<code>1s</code>	WAL-poll fallback + safety-net reconcile interval (<code>0</code> disables polling, <code>fsnotify</code> only)

§4.3 Environment

Variable	Used by	Meaning
<code>HERALD_WORKABLE_DB</code>	<code>pherald watch</code>	DB path fallback when <code>--db</code> is unset
<code>HERALD_CHANNELS</code>	channel setup (shared with <code>pherald listen</code>)	Comma-separated enabled channels (e.g. <code>telegram</code>)
per-channel namespaced <code>env</code>	channel setup	Credentials + target per channel (e.g. the Telegram bot token + target chat id). See <code>docs/guides/MESSENGER_CHANNELS.md</code> §2–§4 and <code>docs/guides/OPERATOR_CREDENTIALS.md</code> .
<code>HERALD_PROJECT_NAME</code>	<code>dispatch envelope</code>	The Herald project name; for the ATMOSphere deployment, <code>ATMOSphere</code> .

The MTProto user-account credentials used by the live test harness (`qaherald`) are configured in the `ATMOSphere.env` per `docs/guides/OPERATOR_CREDENTIALS.md`; they are a test-driver bootstrap, not a `pherald watch` runtime dependency.

§4.4 What subscribers receive — current fan-out caveat

`pherald watch` derives its recipient set from the per-channel configured `Target` (the chat/channel id), so it notifies the operator channel directly. Full PG-backed `Subscriber` resolution is **PENDING (HRD-156)**; until it lands, watch fans out to the configured channel targets, mirroring the explicit-recipient bypass that `workflow.NewNotifier` documents. Preference / quiet-hours filtering is **PLANNED (HRD-154)** and is not yet applied.

Subscribers receive one message per change, using the formats in §6 ( created,  status,  field,  content,  removed).

§5. Inbound actions

Inbound runs under `pherald listen` (the existing Wave 6 runtime). The operator message is dispatched to Claude Code, whose reply must contain a `<<<HERALD-REPLY>>>` block followed by a JSON object. `ParseReply` extracts it into a typed `Reply`; `Action` defaults to "reply" when omitted. A missing marker or malformed JSON is an explicit error — never a fabricated reply (§107 anti-bluff).

§5.1 The `<<<HERALD-REPLY>>>` schema (workable actions)

`item.update` — apply `column→value` deltas to one item:

```
<<<HERALD-REPLY>>>
{
  "action": "item.update",
  "item_update": {
    "atm_id": "ATM-238",
    "location": "Issues",
    "fields": { "status": "Ready for testing", "severity": "Medium" }
  }
}
```

Updatable fields: `type`, `status`, `severity`, `title`, `description`, `forensic_anchor`, `closure_criteria`, `composes_with`, `body_md`, `last_modified`. The composite-key columns (`atm_id`, `current_location`) are NOT updatable in place — a move between locations is a delete + create. An unknown field name is rejected loudly.

`item.delete` — remove one item by composite key:

```
<<<HERALD-REPLY>>>
{
  "action": "item.delete",
  "item_delete": { "atm_id": "ATM-238", "location": "Issues" }
}
```

`investigation.start` — gather info, return a report, optionally propose ONE machine-executable mutation (deferred behind confirmation):

```
<<<HERALD-REPLY>>>
{
  "action": "investigation.start",
  "investigation": {
    "topic": "Why is ATM-238 still failing on D3?",
    "proposed_actions": ["Re-run the Netflix login flow", "Capture logcat"],
    "proposed_action": {
      "kind": "update",
      "atm_id": "ATM-238",
      "location": "Issues",
      "fields": { "status": "Reopened" }
    }
  }
}
```

When `proposed_action` is omitted, the investigation is report-only — no confirmation prompt, no pending action.

§5.2 The act-with-confirmation flow

1. `investigation.start` with a `proposed_action` records the proposal in the `pendingStore` under a token and replies with a report ending in: `Reply CONFIRM <token> to apply: <kind> <atm_id>/<location>`
2. The operator replies `CONFIRM <token>` (the `CONFIRM` keyword is case-insensitive; the token is the next whitespace-delimited field, taken verbatim).
3. The dispatcher takes (and consumes) the pending proposal and executes it via the `ItemMutator` (Update or Delete), then replies `Applied: <kind> <atm_id>/<location>`.

Safety properties (verified): an unknown / already-consumed token is an explicit error (no fabricated success); consuming on `take` means a replayed `CONFIRM` cannot double-apply; if no `ItemMutator` is configured the path returns an explicit error rather than silently succeeding.

§6. Notification message formats

`workflow.RenderChange` produces one deterministic single-line message per `Change`. The renderer never panics or returns empty — an unknown `Kind` falls back to "`<atm_id> <kind>`".

Kind	Constant	Example rendered line
Item created	<code>item.created</code>	 ATM-238 created
Status changed	<code>item.status.changed</code>	 ATM-238 status: In progress → Ready for testing
Field changed	<code>item.field.changed</code>	 ATM-238 severity: Critical → Medium
Content updated	<code>item.content.updated</code>	 ATM-238 content updated
Item removed	<code>item.deleted</code>	 ATM-238 removed

The diff engine (`commons_workable.Diff`) classifies a status difference as `item.status.changed`; `severity / title / type` differences each emit one `item.field.changed`; `body_md / description` differences each emit one `item.content.updated`. Output is deterministically ordered (by `atm_id`, then `current_location`, then a `Kind` rank, then field name). Each `Change` also maps 1:1 to a `CloudEvent` via `workflow.ChangesToEvents`: type `digital.vasic.herald.workable.<kind>`, subject `item:<atm_id>`, JSON body `{atm_id, location, field, old, new}`, fresh UUIDv7 id.

The status-summary line in this guide's header calls these "Jira/ClickUp-style" diff lines. The current renderer emits the single-line forms above; richer formatting polish (multi-field grouping, attribution by/on) is tracked under HRD-151.

§7. Testing and evidence

§7.1 Hermetic coverage that exists today (LIVE, green)

All of the following pass under `go test`:

```
go test -count=1 ./commons_workable/... ./commons_watch/... \
    ./pherald/internal/workflow/... ./pherald/
    internal/inbound/...
```

- `commons_workable` — real SQLite (modernc.org/sqlite, temp DB):
TestOpen_CreatesSchemaIdempotently, TestCRUD_RoundTrip,
TestCreate_RejectsUnknownStatus,
TestCreate_RejectsEmptyStatus, TestUpdate_LoudOnMissing,
TestDelete_LoudOnMissing; diff: TestDiff_Created/Deleted/
StatusChanged/FieldChanged_TitleSeverityType/ContentUpdated/
DeterministicOrderingAcrossItems/NoChange; parser:
TestParseTracker_RepresentativeItem/PlainPrefixItem/
SectionHeaderSkipped/DerivesStableIDWhenNoBracket/ItemCount.
- `commons_watch` — real fsnotify watcher + real files:
TestWatch_EmitsOnModify, TestWatch_DebounceCoalesces,
TestWatch_PollFallbackDetectsSidecar,
TestWatch_CancelNoGoroutineLeak.
- `pherald/internal/workflow` — TestChangesToEvents,
TestRenderChange, TestNotifier_FeedsRealDispatcher (drives the real
runner.ChannelDispatcher through a recording commons.Channel sink — no
mock of the bridge itself).
- `pherald/internal/inbound` — ItemMutator against a real SQLite store
(repo_mutator_test.go), the action router with a recording fake, and the
investigation defer / confirm-executes / replayed-confirm-rejected assertions.
- `pherald_watch` — watch_test.go drives the real runWatch loop (real temp DB,
real fsnotify watcher, real Diff, real Notifier over a recording channel) and asserts a
real DB mutation produces a real rendered diff message through the real fan-out. The
watchDeps.Ready channel is a startup-ordering signal only (closed after baseline +
watcher start), not a pipeline mock.

These are anti-bluff in posture: the PASS bar is "a real DB mutation produces a real rendered diff message dispatched through the real fan-out", not "the process boots".

§7.2 Where evidence lands

Per §107.x, every shipping feature lands a `docs/qa/<run-id>/transcript`. The
watch→notify and inbound→CRUD features land their evidence under `docs/qa/HRD-NNN-
<TS>/` in the same logical work effort.

§7.3 What is PENDING

- **Live Telegram round-trip via MTProto (HRD-156).** A real SQLite mutation → real Telegram message captured by the qaherald MTProto user-account (WaitForReply), plus an inbound command → real row mutation, plus exact byte-for-byte diff-payload assertions. This requires the operator credential bootstrap (`.env` + a one-time qaherald

mtproto login). Until then these live tests honestly SKIP-with-reason (§11.4.3) — they do not PASS.

- **Stress + chaos (§11.4.85) and the paired §1.1 mutation gate** for the integration path (tests/test_atmosphere_integration_mutation_meta.sh) are **PENDING (HRD-156)**. There is currently no tests/test_*/atmosphere* / tests/test_*/workable* shell gate in the repo.
 - **HelixQA / Challenges registration** on the ATMOSphere side is **PLANNED (Phase 3 / HRD-157)**.
-

§8. Setup checklist and troubleshooting

§8.1 Operator pre-deploy checklist



Constitution discoverable from this checkout (tests/test_constitution_inheritance.sh green).



PENDING: docs/workable_items.db materialized + committed (HRD-155 / HRD-131 Phase 3). Until then pherald watch has no items to diff.



HERALD_CHANNELS set and the per-channel credentials present (see OPERATOR_CREDENTIALS.md + MESSENGER_CHANNELS.md).



HERALD_PROJECT_NAME=ATMOSphere for the ATMOSphere deployment.



Hermetic tests green: go test -count=1 ./commons_workable/... ./commons_watch/... ./pherald/internal/workflow/... ./pherald/internal/inbound/....



For inbound: pherald listen configured with an ItemMutator bound to the real commons_workable.Repo (else item.update/item.delete/CONFIRM return "no ItemMutator configured").



PENDING: live MTProto evidence captured + committed under docs/qa/HRD-156-<TS>/ before claiming the live round-trip works (HRD-156).

§8.2 Troubleshooting cookbook

Symptom	Likely cause	Fix
pherald watch: open workable DB ...: ...	DB path wrong or directory missing	Check --db / HERALD_WORKABLE_DB; ensure the parent dir exists. Note the DB itself may not be materialized yet (§3.5).
pherald watch: no channels enabled (HERALD_CHANNELS resolved empty)	No channels configured	Set HERALD_CHANNELS and the per-channel env (see MESSENGER_CHANNELS.md).
Mutations don't notify	WAL writes not seen	Ensure --poll > 0 (default 1s); the watcher watches -wal/-shm sidecars and runs a

Symptom	Likely cause	Fix
		safety-net reconcile, but <code>--poll 0</code> disables both.
Notification fires but the diff is wrong/empty	Snapshot vs. parser mismatch	The diff compares <code>Repo.List</code> snapshots; confirm the rows actually changed in <code>items</code> (not just the MD tracker, until the regenerator HRD-150 lands the two can drift).
<code>item.update</code> errors unknown/unupdatable field "X"	Field not in the updatable set	Use a field from §5.1; composite-key columns are not updatable in place.
invalid status "X" (not in closed set)	Status outside the 10-value set	Use a value from §3.3 verbatim (including the (<code>→ Fixed.md</code>) suffix where applicable).
CONFIRM ...: no pending action for token	Token unknown / already consumed / wrong token	Re-run the investigation to get a fresh token; a CONFIRM consumes its proposal once.
<code>action=item.update</code> but no <code>ItemMutator</code> configured	<code>pherald</code> listen started without a mutator	Bind a <code>RepoMutator</code> over the real <code>commons_workable.Repo</code> .
Live Telegram test "fails" / skips	MTPROTO credentials not bootstrapped	Expected — the live suite SKIPS-with-reason until the operator bootstrap (HRD-156). It is not a PASS yet.

§9. References

Master plan

- `~/Documents/ATMOSphere_Herald_Integration_Plan.md` — architecture, phasing, requirements R1–R7, work-stream decomposition, open operator decisions.

Work-items (HRD) — filed in `docs/Issues.md`:

HRD	Scope	Status
HRD-148	<code>commons_workable</code> SQLite SSoT foundation	in_progress (landed + tested)
HRD-149	<code>commons_watch</code> file/DB watcher	in_progress (landed + tested)
HRD-150	bidirectional MD[?]SQLite regenerator + drift resolution	open (PLANNED)
HRD-151	<code>change→CloudEvent→Runner</code> bridge + diff message formatter	open (bridge built; formatter polish open)
HRD-152	inbound action registry + <code>ItemMutator</code> + investigation orchestrator	open (built + tested)
HRD-153	watcher daemon endpoint (<code>pherald watch</code>)	open (built + tested)
HRD-154	preference / quiet-hours routing enforcement	open (PLANNED)
HRD-155	operationalize the workable-items SQLite tool + reconcile HRD-131	open (PENDING — external blocker for SSoT materialization)

HRD	Scope	Status
HRD-156	anti-bluff full-automation test suite (MTPROTO live, stress/chaos, paired mutation, HelixQA)	open (PENDING — credential bootstrap)
HRD-157	ATMOSPHERE Phase-3 registration (tools/herald submodule), materialize DB, wire runner, live evidence	open (PLANNED)
HRD-158	anti-bluff covenant verbatim-phrase propagation to ATMOSPHERE QWEN.md files	open (PLANNED)
HRD-131	migrate text trackers to versioned SQLite SSoT (6-phase)	open (Phase 1 filed; later phases follow-on)

Source paths (Herald, verified)

- commons_workable/{store.go, item.go, crud.go, changefeed.go, parser.go} — SQLite store + schema, Item + status closed set, CRUD repo, per-property Diff, ATMOSPHERE tracker parser.
- commons_watch/watch.go — fsnotify + WAL-poll watcher.
- pherald/internal/workflow/workflow.go — ChangesToEvents, RenderChange, Notifier.
- pherald/internal/inbound/{reply.go, item_mutator.go, pending.go, dispatcher.go} — <<<HERALD-REPLY>>> schema, ItemMutator/RepoMutator, pending/confirm store, action routing.
- pherald/cmd/pherald/watch.go — the pherald watch daemon (runWatch loop, withWALSidcars, snapshot).

Related guides

- docs/guides/MESSENGER_CHANNELS.md — channel framework, HERALD_CHANNELS, per-channel config, inbox, self-filter.
- docs/guides/OPERATOR_CREDENTIALS.md — credential setup for every messenger + dispatcher (Telegram, Claude Code, MTPROTO bootstrap).

Sources verified

This guide documents Herald's own committed source code and the in-repo master plan; all claims were cross-referenced against the code paths cited in §9 on 2026-05-29. Per §11.4.99, no external-service instructions are introduced here that are not already covered (and source-verified) by MESSENGER_CHANNELS.md / OPERATOR_CREDENTIALS.md.

- Herald source tree (read 2026-05-29): commons_workable/, commons_watch/, pherald/internal/workflow/, pherald/internal/inbound/, pherald/cmd/pherald/watch.go.
- ~/Documents/ATMOSPHERE_Herald_Integration_Plan.md (rev 1, 2026-05-29).
- docs/Issues.md (HRD-131, HRD-148..HRD-158 rows, read 2026-05-29).

Verified 2026-05-30: internal doc — no external online sources. All claims derive from in-repo Herald source (commons_workable/, commons_watch/, pherald/internal/workflow/, pherald/internal/inbound/, pherald/cmd/pherald/watch.go) and the cited in-repo trackers; any external-service instructions are deferred to MESSENGER_CHANNELS.md / OPERATOR_CREDENTIALS.md, which carry their own §11.4.99 source-verification footers. Re-verify on Herald source breaking changes (workflow/inbound API edits, SQLite schema bumps).